The `natex-pdflatex` package

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Abstract

A collection of commands focused on consistent notation for mathematics, physics, and engineering. The repository for this package can be found at: `https://github.com/amilkyboi/natex`.

Contents

1 Included packages

This package requires the `amssymb`, `bm`, `isomath`, `mathtools`, and `upgreek` packages.

2 Commands

2.1 Automated bracing

Command	Usage	Output	Definition
`\abs`	`\abs{x}`	$\lvert x\rvert$	absolute value
`\norm`	`\norm{x}`	$\lVert x\rVert$	norm
`\comm`	`\comm{x}{y}`	$[x, y]_-$	commutator
`\acomm`	`\acomm{x}{y}`	$[x, y]_+$	anticommutator
`\pb`	`\pb{x}{y}`	$\{x, y\}$	Poisson bracket
`\order`	`\order{x}`	$\mathcal{O}(x)$	order of magnitude
`\eval`	`\eval{x}`	$x\rvert$	evaluation limits

2.2 Vector notation

Command	Usage	Output	Definition
`\vb`	`\vb{x}`	$\boldsymbol{x}$	bold vector
`\vu`	`\vu{x}`	$\hat{\boldsymbol{x}}$	unit vector
`\matx`	`\matx{A}`	$\boldsymbol{A}$	matrix
`\tens`	`\tens{T}`	$\boldsymbol{\mathsf{T}}$	tensor
`\vdot`	`\vb{x}\vdot\vb{y}`	$\boldsymbol{x} \cdot \boldsymbol{y}$	dot product
`\vcrs`	`\vb{x}\vcrs\vb{y}`	$\boldsymbol{x} \times \boldsymbol{y}$	cross product
`\del`	`\del`	$\boldsymbol{\nabla}$	del
`\grad`	`\grad{x}`	$\boldsymbol{\nabla} x$	gradient
`\divr`	`\divr{\vb{x}}`	$\boldsymbol{\nabla} \cdot \boldsymbol{x}$	divergence
`\curl`	`\curl{\vb{x}}`	$\boldsymbol{\nabla} \times \boldsymbol{x}$	curl
`\slap`	`\slap{x}`	$\nabla^2 x$	scalar Laplacian
`\vlap`	`\vlap{\vb{x}}`	$\boldsymbol{\nabla}^2 \boldsymbol{x}$	vector Laplacian
`\dalem`	`\dalem`	$\Box$	d'Alembertian

2.3 Dirac notation

Command	Usage	Output	Definition
`\bra`	`\bra{x}`	$\langle x\vert$	bra
`\ket`	`\ket{x}`	$\vert x\rangle$	ket
`\ev`	`\ev{x}`	$\langle x\rangle$	expectation value
`\ip`	`\ip{x}{y}`	$\langle x\vert y\rangle$	inner product
`\op`	`\op{x}{y}`	$\vert x\rangle\langle y\vert$	outer product
`\mel`	`\mel{x}{y}{z}`	$\langle x\vert y\vert z\rangle$	matrix element
`\rmel`	`\rmel{x}{y}{z}`	$\langle x\Vert y\Vert z\rangle$	reduced matrix element

2.4 Set notation

Command	Usage	Output	Definition
`\N`	`\N`	$\mathbb{N}$	set of natural numbers
`\Z`	`\Z`	$\mathbb{Z}$	set of integers
`\Q`	`\Q`	$\mathbb{Q}$	set of rational numbers
`\R`	`\R`	$\mathbb{R}$	set of real numbers
`\C`	`\C`	$\mathbb{C}$	set of complex numbers
`\set`	`\set{a, b, c}`	$\{a, b, c\}$	set notation
`\set`	`\set{a \given b, c}`	$\{a \mid b, c\}$	set builder notation

2.5 Matrix notation

Command	Usage	Output	Definition
`\pmx`	`\pmx{1 & 2 \\ 3 & 4}`	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	parenthetical matrix
`\bmx`	`\bmx{1 & 2 \\ 3 & 4}`	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	bracketed matrix
`\vmx`	`\vmx{1 & 2 \\ 3 & 4}`	$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$	vertical matrix
`\cmx`	`\cmx{1 & 2 \\ 3 & 4}`	$\begin{Bmatrix} 1 & 2 \\ 3 & 4 \end{Bmatrix}$	curly matrix
`\tr`	`\tr{\matx{A}}`	$\operatorname{tr} \boldsymbol{A}$	trace
`\adj`	`\adj{\matx{A}}`	$\operatorname{adj} \boldsymbol{A}$	adjugate
`\tp`	`\matx{A}^\tp`	$\boldsymbol{A}^\mathsf{T}$	transpose
`\cc`	`\matx{A}^\cc`	$\boldsymbol{A}^*$	complex conjugate
`\hc`	`\matx{A}^\hc`	$\boldsymbol{A}^\dagger$	Hermitian conjugate

2.6 Linear operators

Command	Usage	Output	Definition
`\sop`	`\sop{x}`	$\hat{x}$	scalar operator
`\vop`	`\vop{x}`	$\hat{\boldsymbol{x}}$	vector operator

2.7 Probability

Command	Usage	Output	Definition
`\erf`	`\erf{x}`	$\operatorname{erf} x$	error function
`\erfc`	`\erfc{x}`	$\operatorname{erfc} x$	complementary error function

2.8 Trigonometric functions

Command	Usage	Output	Definition
`\asin`	`\asin{x}`	$\operatorname{asin} x$	arcsine
`\acos`	`\acos{x}`	$\operatorname{acos} x$	arccosine
`\atan`	`\atan{x}`	$\operatorname{atan} x$	arctangent
`\asec`	`\asec{x}`	$\operatorname{asec} x$	arcsecant
`\arcsec`	`\arcsec{x}`	$\operatorname{arcsec} x$	arcsecant
`\acsc`	`\acsc{x}`	$\operatorname{acsc} x$	arccosecant
`\arccsc`	`\arccsc{x}`	$\operatorname{arccsc} x$	arccosecant
`\acot`	`\acot{x}`	$\operatorname{acot} x$	arccotangent
`\arccot`	`\arccot{x}`	$\operatorname{arccot} x$	arccotangent
`\sech`	`\sech{x}`	$\operatorname{sech} x$	hyperbolic secant
`\csch`	`\csch{x}`	$\operatorname{csch} x$	hyperbolic cosecant

2.9 Mathematical constants

Command	Usage	Output	Definition
`\img`	`\img`	i	imaginary unit
`\eul`	`\eul`	e	Euler's number
`\pie`	`\pie`	π	pi

2.10 Other

Command	Usage	Output	Definition
`\real`	`\real`	Re	real part
`\imag`	`\imag`	Im	imaginary part
`\defas`	`\defas`	$\coloneqq$	defined as